

William F. Nuveen

San Francisco, CA • wnuveen3@gmail.com • [linkedin.com/in/williamnuveen](https://www.linkedin.com/in/williamnuveen)
williamnuveen.github.io/portfolio/

SUMMARY

Robotics design mechanical engineer with 6+ years of experience across cell-therapy manufacturing, warehouse automation, and consumer robotics. I specialize in designing injection-molded & machinable consumables and hardware interfaces for robotic systems, with a focus on mass manufacturability and reliable human/robotic operation.

PROFESSIONAL EXPERIENCE

MULTIPLY LABS INC.

San Francisco, CA

Senior Robotics HW Engineer - Consumables R&D

May 2026 - Present

- Lead consumables designer for robotic automation of the Rotea cell-therapy instrument; designed the cartridge, consumable, and hardware interface that enable autonomous, end-to-end operation of the Rotea by a robotic arm
- Lead designer for Multiply Labs' pharmaceutical-grade (GMP) bag cartridge across Alpha, Beta, and Gamma revisions, improving reliability, robotic pick-and-place success rate, and regulatory manufacturability while reducing customer assembly time
- Redesigned a peristaltic pumping bag cartridge attachment, cutting per-use cost from \$100 to \$2.50 while simplifying assembly and improving cell-pumping performance
- Provide consumables and hardware support for robotic cluster deployments and operation across 3 large cell-therapy customers
- Managed a summer mechanical engineering intern, defined multiple project scopes, set goals, and tracked progress — providing day-to-day mentorship and technical direction

Advanced Robotics HW Engineer - Consumables R&D

Dec. 2024 - May 2026

- Second hire on Multiply Labs' consumables design team; designing injection molded single-use and reusable pharmaceutical-grade consumables for robotic pharmaceutical manufacturing
- Led Multiply Labs' largest customer pilot to date — 6 cell-therapy batches/day, 20 hrs/day, over 12 continuous days — achieving a 100% batch success rate
 - Owned the full consumables and hardware stack supporting the pilot, including cartridge/consumable manufacturing, robot builds, cluster upgrades, and hardware/testing management, keeping the system production-ready for the pilot's full duration
- Led design, fabrication, and testing of calibration and leveling jigs, enabling bring-up and calibration of 4 robotic clusters and accurate pick-and-place operations across the cluster fleet
- Supported development, deployment, and operation of Multiply Labs' first customer cell-therapy manufacturing robotic system for Legend Biotech

OCADO TECHNOLOGY (FORMERLY KINDRED AI)

San Francisco, CA

Mechanical Engineer II - Robotics Product Development

Sept. 2022 - Dec. 2024

- Core contributor to the end-to-end design and development of Ocado's first AI/ML-driven robotic depalletizing system, deployed at multiple European fulfillment centers
- Increased automated robotic pick-and-place accuracy from 75% to 98% through strategic selection and implementation of depth/RGB cameras, boosting system throughput by 30%
- Co-designed and fabricated an electromechanical end-effector using sensors, actuators, and pneumatics, increasing payload capacity to 80 lbs and improving item-handling versatility

- Conducted load calculations and structural FEA analysis for securing a 1,250 lb robotic arm to a pedestal on concrete, specifying anchors and adhesives for 100% safety compliance and reliable installation
- Managed cross-functional installation and integration of the San Francisco proof-of-concept depalletizing test cell, which led to securing funding approval for the program
- Co-owned hardware design of a portable trade show model for the robotic depalletizing system, enhancing brand visibility

NEATO ROBOTICS

San Jose, CA

Jr. Mechanical Engineer - Product Design

Aug. 2020 - Sept. 2022

- Designed, prototyped, and mass-manufactured injection-molded parts for Neato's first commercial robotic vacuum (Nexaro NR1500) now sold globally by Vorwerk
- Redesigned an electromechanical robotic vacuum drivetrain, improving motor efficiency, reducing noise levels by 30%, and cutting production costs by \$2 per unit
- Streamlined Neato Robotics' additive manufacturing operations, reducing design iteration time by 75% and saved \$20,000 in vendor costs annually
- Developed and scaled high-volume plastic robotic vacuum accessories (pet brush, fragrance dustbin filter), expanding the product line by 7 SKUs
- Initiated and led Neato Robotics' emerging technologies R&D program, developing a novel dust-sensing technology using accelerometer and gyroscopic data to enable adaptive suction control

EDUCATION & CERTS.

MIAMI UNIVERSITY

Oxford, OH

B.S. in Engineering, Major in Mechanical Engineering

2016-2020

GD&T CERTIFICATION - [2-B Quality - ASME Y14.5-2009]

Toronto, ONT

TECHNICAL SKILLS

- **Design:** Onshape (Multiply Labs), Solidworks (Ocado), Creo (Neato), Injection Molding, DFM&A, GD&T, RCA, FAI
- **Building:** Additive Manufacturing, Subtractive Manufacturing (Machining), Injection Molding, Laser Cutting, Sheet Metal, Milling
- **Project Management:** Arena PLM, Confluence, Jira, BOMs, Vendor Management